AGU #774

The Latest on the Reconstruction of the Sunspot Number

Laure Lefèvre[1], W. Dean Pesnell[2], Chandrashekar Kalugodu[1]

(1) World Data Center SILSO, Royal Observatory of Belgium, Brussels, (2) NASA GSFC

The Sunspot Series: S_N and G_N

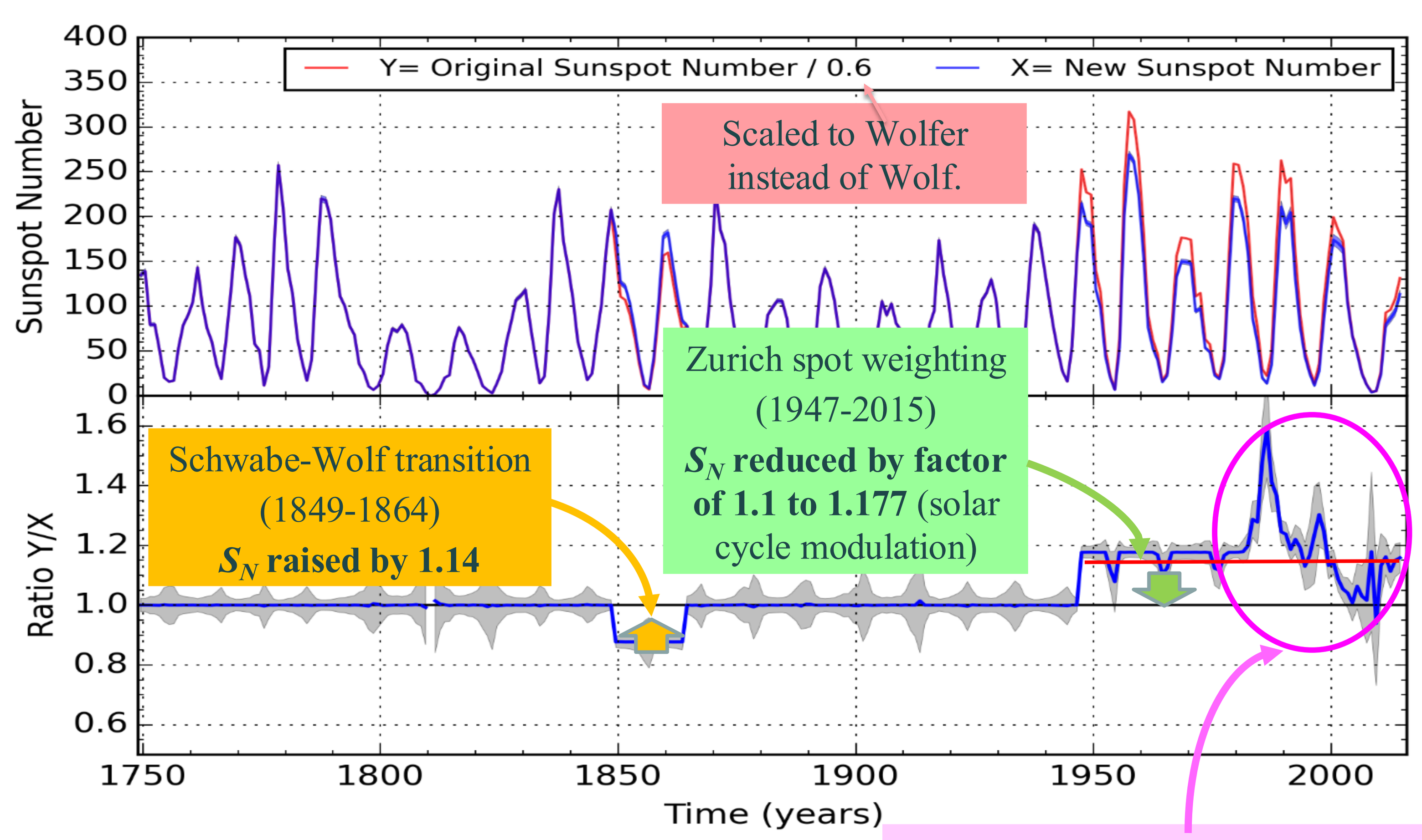
- Visual sunspot counts: longest existing record of solar activity: 400+ years
  - Unique reference for solar dynamo models, climate studies, and solar activity predictions
  - Direct proxy of magnetic flux emergence
- 2 proxy series : Sunspot Number (S_N) and Group Number (G_N)
- SN created in 1849 (R. Wolf, Zurich Observatory):

  S_N= 10 number of groups + number of spots.
- Produced at the World Data Center SILSO in Brussels since 1981:
  - Worldwide network of over 85 stations
  - High (99%) correlation with modern quantitative indices (F10.7, UV flux, spot areas)
- 2010-2015: first ever re-calibration of the entire series. → S_N V2.0
- G_N defined by Hoyt & Schatten (1998):

  G_N = 12.08 x number of groups.
- G_N only source of sunspot counts before 1700.

In 2010 : Revision needed because G_N showed systematic disagreement with S_N before the 20th century.

S_N V2.0: Summary of Corrections



Plot shows the original (red) and new (blue) S_N series (top) and the ratio V1/V2 in bottom panel.

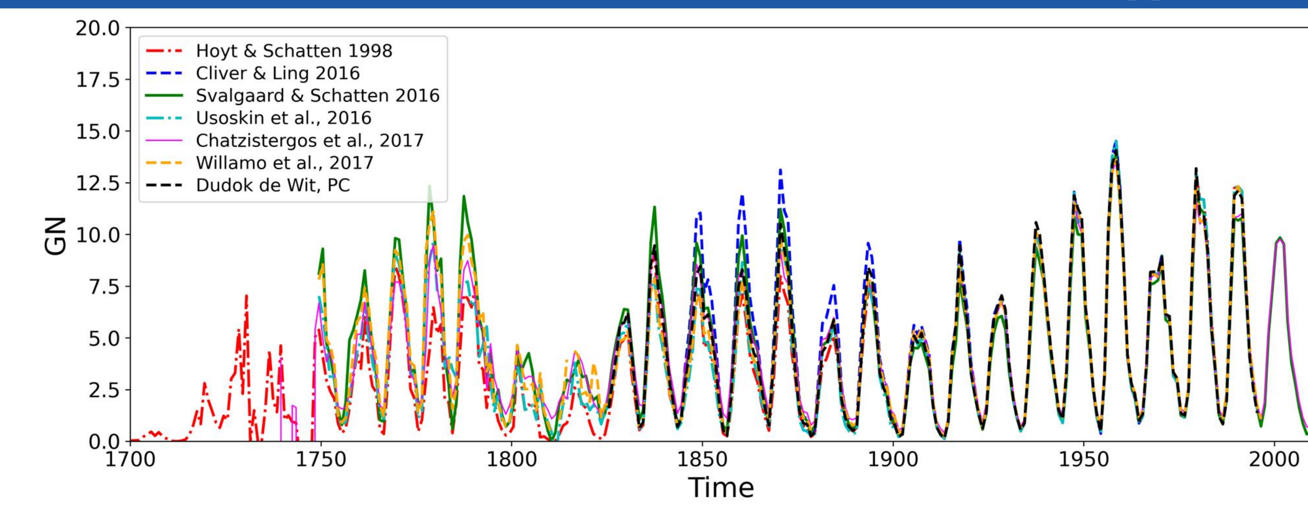
Why is S_N V3.0 Necessary ?

S_N V2.0 is based on a recalibration of the original Sunspot Number (S_N V1.0), i.e. correction factors have been applied to known inconsistencies based on studies and evidence. However, the exact reasons behind those problems were not all known at the time S_N V2.0 was released. For example, the jump of about 20% in 1849 was corrected but the reason for it was discovered later and explained in Bhattacharya et al. (2023, 2024). From these studies, it is evident that a **complete reconstruction from available raw sunspot data (S_N V3.0)** will be **more appropriate than a recalibration using existing data**. Also, error bars for the recalibration were computed based on a statistical study (Dudok de Wit et al., 2016), while the complete reconstruction will make use of the quality of each individual dataset to build error bars.

S_N V3.0 will build on all the historical data that has been gathered thanks to a common effort of all the scientific community and the **FARSUN project** and leverage results to **reconstruct the international sunspot number from raw data and new methods** such as presented in Bhattacharya et al. (2024).

The modern part of the Sunspot Number (1981-present) that is stored in the WDC-SILSO database, will also be completely reconstructed and delivered at the same time as the complete reconstruction of the historical part of the S_N (1610-1980). This aspect of the work is covered by the **SUNRISE project.**

Reconstruction of G_N



After the 2016 release of the GN database (Vaquero et al., 2016), multiple, methodologically distinct reconstructions of G_N were published which mostly disagree over the 19th century (see figure above). The table below from Clette et al. (2023) sorts Sunspot Series in 3 different categories (low, medium, high- depending on their overall level over the 19th century) and assesses which S_N (S_N V1.0 and V2.0) and G_N series correspond to which benchmark series (Clette et al., 2023). The overall answer provided by those comparisons remains partly ambiguous. Further progress is needed to elucidate the remaining disagreements, and this is what the current poster is addressing.

	G_N and S_N series	Observer correction factor (Figure 14)	$S_N(2.0)/G_N$ (Figure 13)	Geomagnetic IDV index (Figure 18)	Diurnal rY modulation (Figure 19)	Isotopes ^{14}C, ^{10}Be (Figure 20)
HIGH	SvSc16 ClLI16 *S_N V2.0*	BEST	BEST	BEST	BEST	NO
MEDIUM	CEA17 DuKo22 UEA21 *S_N V1.0* *LEA14*	FAIR (low)	FAIR (low)	FAIR (low)	FAIR (low)	BEST
LOW	HoSc98	NO	NO	NO	NO	FAIR (low)

From Clette et al. (2023): SvSc16 (Svalgaard and Schatten, 2016); ClLI16 (Cliver and Ling 2016);CEA17 (Chatzistergos et al., 2017);DuKo22 (Dudok de Wit, in prep.); UEA16 and 21 (Usoskin et al., 2016 and 2021), LEA14 (Lockwood et al., 2014)

Do we need G_N V3.0 ?

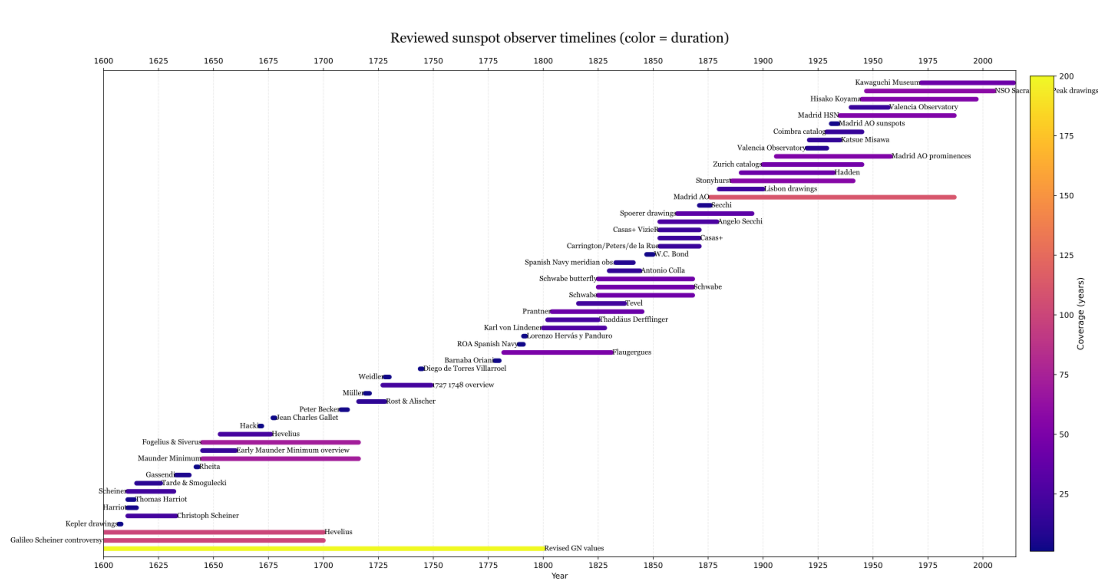
The short answer is a resounding yes! The equivalent of S_N V3.0 for the G_N series is needed. The science community needs a unique, vetted series for solar cycle predictions, space weather operations, climate studies and many other fields. Moreover, the database with group numbers is in constant evolution and a new version is in construction as of 2025. This calls for a consistent study of all the methods whose results are presented above and the homogeneous use of one database in order to construct one G_N series that would be validated and vetted by the Solar Physics community.

In addition to that, the WDC SILSO is working on the future production of G_N at the same rate as S_N is produced today (see SUNRISE project). This new G_N product is set to be released in 2027.

What progress since S_N V2.0 ?

Since the release of SN V2.0 in 2015, ISSI team meetings have been organized and a Sunspot Workshop took place in 2024. The goal is to achieve a complete reconstruction of SN and GN from a well-populated database of raw sunspot data (see FARSUN project). This implies both an important effort at gathering datasets for key periods where information is missing (see figure), but also an effort at analyzing all methods used to combine datasets together (Chatzistergos et al., 2025). We focus here on the data recovery effort. Between 2017 and 2023 the WDC–SILSO digitized the *Mittheilungen* (the Zürich Observatory journals initiated by Rudolf Wolf that contain data from 1610 to 1944) as well as the Zürich observation tables (1945-1979; FARSUN). In parallel, data-recovery efforts were carried out by Arlt, Carrasco, Friedli, Hayakawa,Vaquero and others.

Data recovery

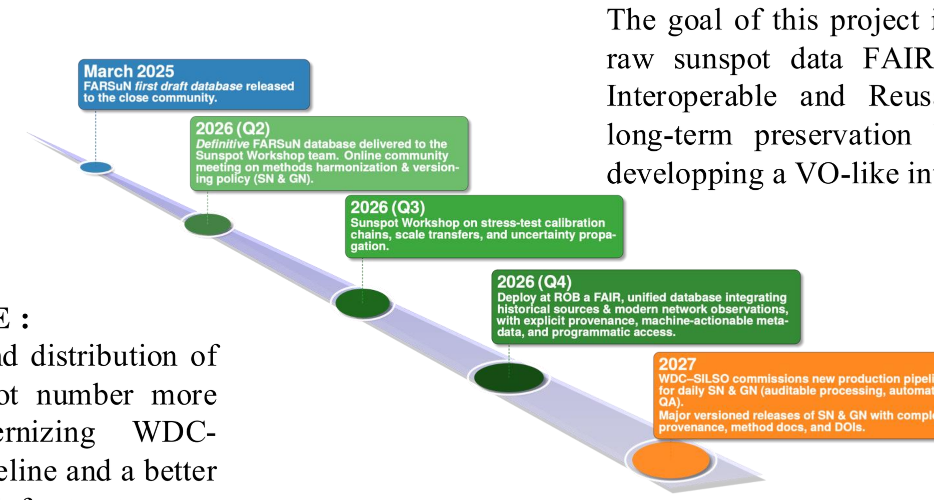


The different teams located, transcribed, and published numerous early telescopic sunspot records spanning from the early 1600s to the modern era. This figure is based on a (non-exhaustive) list of publications on raw sunspot data since 2010. Together, these initiatives expanded the historical observation base, enabling independent cross-checks of observers and providing the foundation for more robust, uncertainty-quantified reconstructions of the long-term sunspot series, mostly by increasing the number of concurrent observations for specific time periods.

When will new S_N and G_N be released ?

FARSUN :

The goal of this project is to make the historical raw sunspot data FAIR (Findable, Accessible, Interoperable and Reusable) and secure their long-term preservation by gathering data and developping a VO-like interface for access.



SUNRISE :

Make the production and distribution of the international sunspot number more sustainable by modernizing WDC-SILSO's production pipeline and a better integration in the SIDC infrastructure.

Conclusions

Since the first ever revision of the international Sunspot Number carried out between 2010 and 2015 and published in 2016 (Clette & Lefèvre, 2016), the community has come together to gather more original sources of observations (Clette et al., 2021; Clette et al., 2023), make them accessible (FARSUN), discuss and use different combination methods (Bhattacharya et al., 2023; Chatzistergos, 2025) and produce community vetted sunspot series that extend to the present (production modernization in SUNRISE).

This all comes together in 2027, when a new database will be released containing all sunspot number and group number data available to reconstruct SN and GN (FARSUN), at the same time as new versions of SN (SNV3.0) and GN are released, in sync with the operational production of GN at SILSO and an improved SN (SUNRISE).

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For sunspot number data and other information visit https://sidc.be/silso

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